A Prediction-Driven Framework for Proactive Pedestrian Red-Light-Running Risk Evaluation

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Abstract

Pedestrian red-light running remains a persistent safety challenge at signalized intersections, where conventional detection-based approaches typically identify violations only after they have occurred. Such retrospective methods provide limited support for proactive safety monitoring because they overlook the dynamic process through which risky crossing intentions emerge, intensify, or are suppressed before observable violations. To address this limitation, this study develops a Hierarchical Prediction-Driven Measurement framework, termed HPDM, with five variants incorporating different intention prediction models, for proactive evaluation of pedestrian red-light-running risk in intelligent intersection environments. Rather than treating prediction as an end in itself, HPDM transforms trajectory-based predictive outputs into interpretable, temporally resolved indicators of latent crossing intention and subsequently models the transition from intention to realized violation behavior. The framework integrates pedestrian trajectories, traffic signal states, crossing region geometry, and surrounding vehicle interactions to quantify how red-light-running propensity evolves under dynamic traffic conditions. Experiments are conducted using a real-world video dataset collected at a signalized intersection near the University of Hong Kong, comprising 30,231 pedestrian trajectories, 36,791 vehicle trajectories, and 1,390 confirmed red-light-running events. Results show

that HPDM (FlowChain) achieves an overall accuracy of 96.74%, with a false alarm rate of 3.18% and a missed detection rate of 5.29% on the collected dataset. On a balanced test set, it maintains an accuracy of 95.80% and achieves a PR-AUC of 0.9643. Comparisons across five prediction backbones demonstrate the framework's flexibility, with alternative implementations also providing good performance. Evaluation across prediction horizons shows that the highest mean accuracies across the variants are 96.46% at 0.4 s and 87.34% at 1.2 s, with performance declining at longer horizons. These findings support the use of HPDM for short-term anticipatory monitoring. The proposed framework shifts pedestrian red-light-running analysis from post-hoc violation detection toward proactive, process-oriented safety monitoring, offering methodological support for intelligent transportation systems, real-time intersection risk assessment, and pedestrian safety management.

Keywords: Red-Light Running, Intention Prediction, Urban Pedestrian, Behavior Analysis and Understanding

1 Introduction

Pedestrian red-light running is widely considered a common pedestrian violation that can cause serious traffic accidents worldwide. According to the Global Status Report on Road Safety published by the World Health Organization (WHO), pedestrians account for approximately 23% of all road traffic deaths globally, highlighting their disproportionate vulnerability among road users (World Health Organization, 2018). A substantial proportion of these fatalities occur in urban environments, particularly at or near signalized intersections. The WHO further identifies risky pedestrian behaviors, including violations of traffic signals, as key contributors to severe injuries and fatalities in cities. Consistent findings have also been reported in regional safety assessments. For example, the European Road Safety Observatory notes that pedestrian crashes are highly concentrated in urban areas and commonly occur at junctions and crossings, where non-compliant crossing behavior substantially elevates injury risk (European Road Safety Observatory, 2018). Taken together, these findings indicate that pedestrian red-light running is not only widespread, but also intrinsically linked to high-risk interaction environments especially in dense urban settings.

Hong Kong exemplifies this broader global challenge in such a highly urbanized context. In 2023, the city recorded 17,189 road traffic accidents involving 22,269 persons, representing year-on-year increases of 14% in accident cases and 20% in casualties. Pedestrians were the most affected group in fatal outcomes, accounting for 65% of all traffic fatalities, among which 63% involved older adults aged 65 or above. Notably, 31% of all accidents occurred at or near pedestrian crossings, indicating a strong spatial association between pedestrian activity and crash occurrence (Hong Kong Police Force, 2024b). These statistics highlight that signalized crossings, while designed for safety, remain critical locations where pedestrian risk is both concentrated and repeatedly manifested. For intelligent transportation systems, such crossings are also important

sensing and decision-making sites, where video streams, signal-state information, and trajectory data can potentially be transformed into real-time indicators of pedestrian safety risk. However, realizing this potential requires methods that move beyond detecting violations after they occur and instead quantify how risk develops before an unsafe crossing action is fully realized.

Existing countermeasures have often emphasized enforcement after violations are observed. For example, Hong Kong has implemented targeted pedestrian enforcement campaigns, under which pedestrians who fail to comply with traffic signal indications may be subject to legal penalties (Hong Kong Police Force, 2024a). During the month-long operation, police issued 1,374 summonses, approximately 80% of which were related to signal violations. While such measures may deter intentional non-compliance, they largely operate at the outcome level and provide limited understanding of how and when risky crossing behaviors emerge. In particular, they fail to capture the formation and evolution of risk prior to violation, thereby offering little insight into the underlying behavioral mechanisms that lead to pedestrian red-light running (Dumbaugh and Li, 2010; Avineri et al., 2012; Liu and Tung, 2014).

A similar limitation is present in much of the existing research on pedestrian red-light running. Previous studies have have examined pedestrian red-light running through observed crossing outcomes and the factors associated with signal noncompliance (Yung and Lai, 2001; Chen et al., 2017; Sun et al., 2025). Outcome-based studies also examine pedestrian injury severity using observed crash records (Song et al., 2020; Barbour et al., 2024). In related traffic-safety applications, video-based systems identify realized red-light-running violations by tracking vehicles relative to traffic signals and spatial boundaries (Silva et al., 2025). Such approaches are valuable for identifying and summarizing realized violations, but they are inherently retrospective. They provide limited insight into the behavioral process through which risk emerges, escalates, or is suppressed before a violation becomes observable. As a result, conventional detection methods and post-hoc violation statistics are insufficient for addressing several safety-relevant questions.

In parallel, recent advances in emerging sensing technologies and artificial intelligence have enabled increasingly accurate pedestrian trajectory and intention prediction in complex traffic environments (Yang et al., 2023; Li et al., 2023; Meng et al., 2026). Studies from transportation, intelligent vehicle, and computer vision communities have developed models to predict future pedestrian motion, crossing intention, or interaction behavior using trajectories (Alahi et al., 2016; Li et al., 2020; Huang et al, 2025), visual features (Zhong et al., 2023; Xia et al. 2026), and contextual information (Xia et al. 2022;Zhang et al., 2023). These models (Wong et al., 2024; Bae et al., 2025; Wong et al., 2025) have significantly improved prediction performance and have demonstrated the value of data-driven methods for understanding pedestrian movements. However, prediction has often been treated as the final objective, with evaluation centered on forecasting accuracy rather than transportation safety interpretation. Consequently, the rich

probabilistic information embedded in predictive outputs has not been fully translated into operationally meaningful risk indicators for intersection monitoring or pedestrian safety management.

This gap is particularly important because severe pedestrian crashes are rare, while risk-related behavioral processes occur much more frequently and can provide earlier safety-relevant signals. A pedestrian may show a rising tendency to cross during a red signal, approach the curb, hesitate, slow down, or stop when encountering vehicle movement. These intermediate behavioral states may not always result in an observed violation, but they are critical for understanding how risk is formed and regulated in real traffic environments. From a transportation-system perspective, such process information is valuable because it can support proactive warning, adaptive signal management, targeted safety diagnosis, and evaluation of intersection design or operational conditions. A method that only detects completed violations cannot provide this type of early and temporally resolved safety evidence.

Therefore, motivated by the principle of "*prevention is better than cure*", this study proposes a **H**ierarchical **P**rediction-**D**riven **M**easurement framework, termed HPDM, for analyzing pedestrian red-light-running propensity. The primary contribution of this study lies in a modular pipeline that connects probabilistic trajectory prediction with continuous behavioral measurement and violation identification. Rather than using prediction as an end in itself, the proposed framework reformulates it as an intermediate step within a structured risk quantification process. The framework integrates historical pedestrian trajectories, traffic signal states, crossing region geometry, and surrounding vehicle conditions into a unified scene representation. Specifically, the framework comprises two stages. First, it estimates future trajectory distributions and computes crossing region probabilities at successive future time steps, providing interpretable, temporally resolved indicators of crossing propensity under red-light conditions. Second, it applies an empirically calibrated threshold to support discrete violation decisions when required, while retaining the continuous probability profiles for behavioral analysis.

These stages are connected through crossing region probabilities as a common intermediate representation, allowing different prediction models to be incorporated while retaining the same downstream measurement and decision-rule structure. We implement five HPDM variants to examine this flexibility, with the prediction models serving as backbone components and thresholding as an interpretable decision rule. By shifting from outcome-based descriptions to a prospective, process-oriented measurement paradigm, the proposed framework extends violation analysis beyond whether a violation occurs to how predicted crossing propensity varies over time and across traffic contexts.

2 Related work

2.1 Outcome-Based Analysis of Pedestrian Red-Light-Running Risk

Existing studies (Yagil, 2000; Yang et al., 2006; Zhou and Horrey, 2010; Dumbaugh and Li, 2010) commonly examine whether pedestrians comply with a red signal and which factors influence the crossing outcome. Field observations show that red-light crossing is associated with gender and the number of pedestrians waiting at the curb (Rosenbloom, 2009), as well as situational conditions such as group size and parked vehicles (Dommes et al., 2015). Using survey data, Zhang et al. (2016) identify trip purpose, time of day, and attitudes toward hurried crossing and road facilities as factors associated with red-light running. Marisamynathan and Vedagiri (2018) combine video observations with a perception survey to model crossing behavior and safety at signalized intersections, and report time saving and convenience as reasons for noncompliance. Raoniar and Maurya (2022) further examine social influences, finding that pedestrians waiting at the intersection and nearby pedestrians crossing on red relate to violation likelihood in opposite directions. Using a stated preference survey, Zhu et al. (2021) examine how the trade-off between time savings and safety risk relates to pedestrians' red-light-running choices.

While these works clarify the factors associated with red-light running, their main analytical outcomes are observed violations or self-reported crossing behaviors. By focusing exclusively on realized red-light-running events, outcome-based approaches are unable to capture how pedestrian risk emerges and evolves prior to observable violations. In particular, they provide limited insight into pedestrians' latent decision-making processes and intention evolution. They also overlook near-violation situations where risky behavior does not ultimately materialize. As a result, these approaches primarily support retrospective assessment and enforcement, rather than enabling proactive understanding or early identification of pedestrian risk.

2.2 Prediction-Based Pedestrian Behavior Analysis and Understanding

In parallel with outcome-based studies, current research (Golchoubian et al., 2023; Wong et al., 2024; Li et al., 2025; Song et al., 2026) in both intelligent transportation and computer vision communities investigates agents' motion using prediction-based models. The pioneering work Social-LSTM (Alahi et al., 2016) models social interactions among nearby pedestrians. Moreover, Lin et al. (2024) learn short-term motion and long-term destination information to predict future trajectories, and Xia et al. (2025) develop a spectral approach for heterogeneous trajectories. Recent works introduce additional context. Bae et al. (2025) use a multimodal language model to reason about scenes and social relationships, whereas Liu et al. (2026) predict pedestrian trajectories from an ego-centric perspective in pedestrian–vehicle mixed environments. For crossing scenarios without traffic signals, Qin et al. (2026) incorporate personalized perceived risk into pedestrian crossing trajectory prediction.

Recent advances in data-driven models have significantly improved prediction performance. However, these improvements are primarily reflected in performance gains on prediction benchmarks rather than leveraged for behavioral analysis. Consequently, prediction is largely treated as an end goal rather than an analytical tool, and the rich information embedded in predictive outputs remains underutilized for behavioral measurement and quantitative risk characterization. As a result, the potential of prediction models for understanding pedestrian risk remains largely underexplored beyond performance evaluation, limiting their applicability for pedestrian risk analysis (Liu et al, 2023; Zhang et al, 2024; Gao et al, 2026).

2.3 Research Gap

Overall, existing studies on pedestrian red-light running mainly follow two research paradigms. On the one hand, outcome-based approaches focus on violation detection and post-hoc statistical analysis, providing descriptive summaries of observed red-light-running events. On the other hand, prediction-based approaches focus on forecasting pedestrian trajectories or intentions, with primary emphasis on predictive performance.

However, a key limitation remains the absence of a unified analytical framework that systematically connects predictive outputs with interpretable behavioral measurement. In particular, there is a lack of methods that transform predictive information into quantitative measures of pedestrian risk, capable of characterizing how pedestrian risk forms, evolves, and may be suppressed prior to observable violations. Addressing this gap calls for a prediction-driven measurement framework that transforms predictive outputs into interpretable risk quantification. Such a paradigm moves beyond both outcome-based analysis and prediction as an end goal, enabling proactive pedestrian risk assessment.

3 Methodology

3.1 Problem Definition

This study focuses on the analysis of pedestrian red-light-running risk in urban traffic environments. Specifically, we aim to characterize and predict whether a pedestrian will enter a designated crossing region while the traffic signal is red, which constitutes a red-light-running event.

At each time step t, the observed scene is represented as:

$$\mathcal{S}_t = \{\mathbf{X}_t, \mathbf{L}_t, \mathbf{R}, \mathbf{C}_t\}, \tag{1}$$

where $\mathbf{X}_t$ denotes the observed pedestrian trajectory, $\mathbf{L}_t$ represents the traffic signal state, $\mathbf{R}$ denotes the predefined spatial crossing region, and $\mathbf{C}_t$ captures contextual environmental factors such as surrounding vehicle movements.

Given the observed scene $\mathcal{S}_t$, the goal is to infer the pedestrian's future behavioral state. Rather than directly estimating a binary outcome, we formulate this process in a hierarchical manner by explicitly modeling the latent crossing intention and its realization into observable behavior:

$$\mathcal{S}_t \xrightarrow{f_{\text{intent}}} P_{\text{cross}}(t+1) \xrightarrow{f_{\text{map}}} \hat{y}_{t+1}, \tag{2}$$

where the intention prediction function f_{intent} estimates a continuous crossing region probability $P_{\text{cross}}(t+1)$, reflecting the pedestrian's latent intention to cross at the next time step, and the mapping function f_{map} transforms this probability into a binary behavioral outcome $\hat{y}_{t+1} \in \{0,1\}$. Here, $\hat{y}_{t+1} = 1$ indicates that the pedestrian enters the crossing region under a red signal, a realized red-light-running event.

This hierarchical formulation not only enables behavior prediction, but also provides a principled basis for quantifying the formation and evolution of red-light-running risk.

3.2 Overall Framework

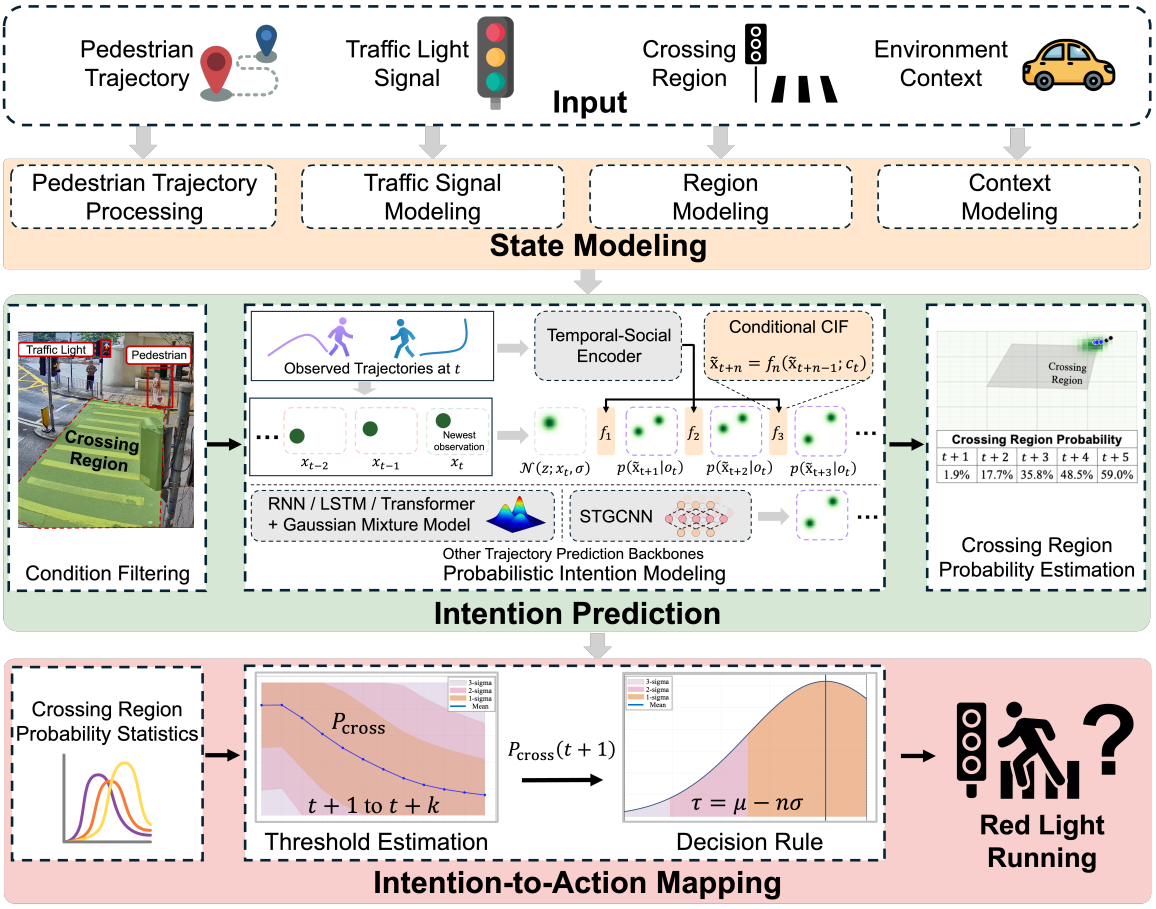


Figure 1. Illustration of the proposed hierarchical prediction-driven measurement framework. The framework processes multimodal input data to predict pedestrian trajectories and progressively maps motion intentions to red-light-running risks.

The architecture of the proposed prediction-driven measurement framework, HPDM, is illustrated in **Figure 1**. The framework takes historical pedestrian trajectories, traffic signal states, crossing regions, and environmental contexts as inputs, and models red-light-running risk in a hierarchical and process-oriented manner.

Specifically, it consists of three key components: (1) **State Modeling** transforms raw observations into a structured representation of the traffic environment for subsequent behavior inference. (2) **Intention Prediction** estimates the pedestrian's latent crossing intention in a probabilistic manner based on historical motion patterns and contextual interactions. (3) **Intention-to-Action Mapping** characterizes the realization of intention into observable behavior by converting continuous intention probabilities into discrete behavioral outcomes.

Unlike conventional prediction pipelines that focus solely on final outcomes, the proposed framework explicitly models the intermediate intention formation process, thereby enabling an interpretable quantification of how red-light-running risk emerges and evolves over time.

3.2.1 State Modeling

The state modeling module aims to construct a structured and behavior-relevant representation of the dynamic traffic environment, serving as the foundation for subsequent intention estimation and behavior inference. Given raw observations, we integrate pedestrian historical motion, traffic signal, spatial region, and surrounding vehicle dynamics into a unified and decision-relevant scene representation.

Pedestrian Trajectory Processing. We first process raw pedestrian trajectories to obtain temporally aligned historical observations. For each pedestrian, an object detection and tracking model (Redmon et al., 2016) is employed to extract trajectory segments with an observation horizon of H, denoted as $\mathbf{X}_t = \{x_{t-H+1}, \dots, x_t\}$, where $x_i \in \mathbb{R}^2$ represents the spatial position at time i. Trajectories that do not satisfy the required observation horizon are discarded to ensure temporal consistency.

To address missing or noisy observations, a two-stage refinement strategy is adopted. Linear interpolation is first applied to fill intermediate missing values, while boundary missing segments are estimated using slope-based extrapolation derived from local temporal gradients. Subsequently, Savitzky-Golay filtering (Savitzky and Golay, 1964) is applied to smooth trajectories, reducing measurement noise while preserving motion trends.

Traffic Signal Modeling. We employ an object detection model to determine the traffic signal state at each time step t, denoted as $\mathbf{L}_t \in \{0,1\}$. Here, $\mathbf{L}_t = 1$ indicates a red-

light interval, while $\mathbf{L}_t = 0$ represents a non-red state. The traffic signal serves as a critical regulatory condition for defining red-light-running events and is temporally synchronized with pedestrian trajectories.

Region Modeling. The spatial extent of the pedestrian crossing is represented as a configurable geometric quadrilateral $\mathbf{R} \subset \mathbb{R}^2$. This region can be flexibly defined according to scene geometry, such as zebra crossing boundaries, enabling generalization across different intersections. A point-in-polygon test is employed to determine whether a pedestrian position lies within $\mathbf{R}$, providing a clear spatial criterion for identifying crossing behavior.

Context Modeling. To capture environmental influences on pedestrian behavior, we model surrounding vehicle dynamics as contextual factors. Let $\mathbf{C}_t$ denote the set of nearby vehicles and their motion states at time t. We focus on actively moving vehicles within a spatial neighborhood of $\mathbf{R}$, as they provide critical interaction cues that may influence pedestrian decision-making. Specifically, vehicle velocity and relative position with respect to the crossing region are incorporated to characterize dynamic interaction conditions.

Overall, the state modeling process transforms raw multimodal observations into a structured representation $\mathcal{S}_t = \{\mathbf{X}_t, \mathbf{L}_t, \mathbf{R}, \mathbf{C}_t\}$, which captures the decision-relevant spatiotemporal context and provides a principled basis for subsequent intention prediction and risk measurement.

3.2.2 Intention Prediction

The intention prediction module aims to estimate the pedestrian's latent crossing intention in a probabilistic manner, rather than predicting deterministic trajectories or discrete behavioral outcomes. Given observed trajectories and environmental contexts, the model characterizes the likelihood of future spatial tendencies that may lead to entering the crossing region during red-light conditions. This probabilistic formulation serves as a critical intermediate layer that explicitly transforms spatial uncertainty into measurable behavioral risk indicators.

Condition Filtering. To ensure that intention estimation is conducted under meaningful and behaviorally relevant scenarios, we introduce a set of prerequisite conditions. First, when the traffic signal indicates a non-red state ($\mathbf{L}_t = 0$), red-light-running is not feasible, and the crossing region probability is directly set to zero. Second, when actively moving vehicles are present in the vicinity of the crossing region (as indicated by $\mathbf{C}_t$), pedestrian crossing behavior is strongly constrained by safety considerations. In such cases, pedestrian crossing intention is assumed to be strongly suppressed and its estimated crossing region probability is reduced accordingly. Third, trajectories with insufficient observations or severe discontinuities are excluded to avoid

unreliable inference. Only samples satisfying all conditions are used for intention estimation.

Probabilistic Intention Modeling. To capture the inherent uncertainty and variability of pedestrian intention, we adopt a flow-based generative framework FlowChain (Maeda and Ukita, 2023) to model the conditional distribution $p(\mathbf{Y}_{t+1:t+K} \mid \mathcal{S}_t)$ of future spatial intention given the current scene $\mathcal{S}_t$.

Instead of predicting a single future path, the model learns a distribution over possible future spatial configurations that reflect potential behavioral tendencies. Specifically, let $z \sim \mathcal{N}(0, I)$ denote a latent variable. A sequence of invertible transformations f_θ maps the latent variable to the intention space conditioned on $\mathcal{S}_t$. Formally,

$$\mathbf{Y}_{t+1:t+K} = f_\theta(z; \mathcal{S}_t). \tag{3}$$

Due to the invertibility of the transformation, the likelihood of the spatial intention distribution can be computed using the change-of-variable formula:

$$p(\mathbf{Y}_{t+1:t+K}) = p(z) \left| \det \left(\frac{\partial f_\theta^{-1}(\mathbf{Y}_{t+1:t+K})}{\partial \mathbf{Y}_{t+1:t+K}} \right) \right|. \tag{4}$$

The flow-based design allows efficient density estimation while preserving expressive modeling of multimodal and uncertain intention distributions.

For each future time step $t + k$, we obtain a marginal distribution over spatial positions. Formally,

$$x_{t+k} \sim \mathcal{N}(\mu_{t+k}, \Sigma_{t+k}), \tag{5}$$

where μ_{t+k} and Σ_{t+k} denote the predicted mean and covariance of the intended position at step $t + k$. These parameters summarize the spatial uncertainty of the intention and reflect the variability of the underlying crossing desire.

Crossing Region Probability Estimation. Based on the estimated spatial intention distribution, we compute the probability that the pedestrian's intended pathway will enter the crossing region **R** at each future time step. Due to the complex geometry of **R**, direct analytical computation of this probability is intractable. Therefore, we adopt a Monte Carlo

sampling strategy. For each time step $t+k$, we draw N samples from the predicted distribution. Formally,

$$\{x_{t+k}^{(i)}\}_{i=1}^{N} \sim p(x_{t+k} \mid \mathcal{S}_t). \tag{6}$$

The empirical probability of entering the crossing region is estimated as:

$$P_{\text{cross}}(t+k) = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left(x_{t+k}^{(i)} \in \mathbf{R}\right), \tag{7}$$

where $\mathbb{1}(\cdot)$ denotes the indicator function.

This sampling-based estimation effectively approximates the integral of the probability density over the crossing region:

$$P_{\text{cross}}(t+k) = \int_{\mathbf{R}} p\left(x_{t+k} \mid \mathcal{S}_t\right) dx. \tag{8}$$

By evaluating $P_{\text{cross}}(t+k)$ over future horizons $k = 1, \ldots, K$, we obtain a temporal profile of crossing intention, which directly characterizes the temporal evolution of red-light-running risk.

Overall, this probabilistic intention modeling framework transforms spatial uncertainty into interpretable risk measures, serving as a key component for subsequent behavior realization and risk analysis.

3.2.3 Intention-to-Action Mapping

While the intention prediction module provides the probability of a pedestrian intending to enter the crossing region, such probability primarily reflects latent crossing desire rather than realized violation behavior. In practice, a non-trivial gap exists between intention and action, as pedestrians may refrain from crossing even when their estimated intention indicates a high likelihood of entering the crossing region. To bridge this gap, we introduce an intention-to-action mapping mechanism that converts probabilistic intention into a discrete behavioral decision.

Threshold Estimation. We learn a data-driven decision threshold to characterize when intention translates into actual red-light-running behavior. Specifically, we collect a

set of crossing probabilities at time $t+1$, denoted as $\{P_{\text{cross}}^{(i)}\}_{i=1}^{M}$, from pedestrians in the training set who are observed to run red lights. These values represent the distribution of intention strength associated with realized red-light-running events.

For each prediction model, we calculate the sample mean μ and standard deviation σ of the crossing-region probability scores associated with confirmed violations in the calibration data. These empirical statistics define the decision threshold as:

$$\tau = \mu - n\sigma, \tag{9}$$

where n controls the position of the threshold relative to the mean. We use $n = 2$, yielding an empirical two-standard-deviation threshold. This rule is calculated directly from the calibration scores and does not assume Gaussianity or guarantee a fixed proportion of retained violation samples. The threshold is calibrated separately for each model and fixed during testing, with its performance assessed using false alarm and missed detection rates.

Decision Rule. During inference, given the predicted crossing region probability $P_{\text{cross}}(t+1)$, the red-light-running behavior is determined as:

$$\hat{y}_{t+1} = \begin{cases} 1, & \text{if } P_{\text{cross}}(t+1) \geq \tau \\ 0, & \text{otherwise} \end{cases}. \tag{10}$$

This rule provides a simple yet effective mapping from probabilistic intention to discrete behavior. Using the same empirical rule across models provides a consistent calibration procedure while allowing the numerical threshold to vary with each model's score distribution.

Overall, the intention-to-action mapping module completes the transition from probabilistic risk estimation to observable behavior realization, enabling accurate and interpretable red-light-running analysis.

3.3 Evaluation Metrics

To quantitatively evaluate the performance of the proposed framework in identifying pedestrian red-light-running behavior, three metrics are adopted: Overall Accuracy, False Alarm Rate, and Missed Detection Rate.

Overall Accuracy. Accuracy measures the overall correctness of the predictions, defined as the ratio of correctly classified samples to the total number of samples:

$$\text{Accuracy} = \frac{N_{\text{TP}} + N_{\text{TN}}}{N_{\text{Total}}}, \tag{11}$$

where N_{TP} and N_{TN} denote the numbers of correctly predicted red-light-running and non-red-light-running samples, respectively, and N_{Total} is the total number of samples.

Missed Detection Rate (MDR). MDR, also known as the False Negative Rate, evaluates the proportion of red-light-running samples that are not correctly detected:

$$\text{MDR} = \frac{N_{\text{FN}}}{N_{\text{RLR}}}, \tag{12}$$

where N_{FN} represents the number of red-light-running samples that are incorrectly predicted as non-violations, and N_{RLR} denotes the total number of red-light-running samples.

False Alarm Rate (FAR). FAR, also known as the False Positive Rate, measures the proportion of non-red-light-running samples that are incorrectly predicted as violations:

$$\text{FAR} = \frac{N_{\text{FP}}}{N_{\text{Non-RLR}}}, \tag{13}$$

where N_{FP} is the number of non-red-light-running samples that are incorrectly classified as violations, and $N_{\text{Non-RLR}}$ is the total number of non-red-light-running samples.

These metrics provide a comprehensive evaluation of the proposed method by jointly considering overall prediction accuracy and error characteristics. In particular, Missed Detection Rate reflects the model's ability to capture risky behaviors, while False Alarm Rate evaluates its robustness in avoiding unnecessary warnings, thereby characterizing the trade-off between safety sensitivity and prediction reliability.

To quantitatively evaluate the reliability of the estimated spatial intention, we adopt two widely used metrics in trajectory analysis: Average Displacement Error and Final Displacement Error. Although originally designed for trajectory prediction, these metrics are employed here to assess how well the inferred spatial intention aligns with the eventually realized motion, thereby evaluating the consistency between intention estimation and actual behavior realization.

Average Displacement Error (ADE) (Pellegrini et al., 2009). ADE measures the average Euclidean distance between the estimated intention positions and the ground-truth observations over all future time steps. Formally, it is defined as:

$$\mathrm{ADE} = \frac{1}{K}\sum_{k=1}^{K} \| \hat{x}_{t+k} - x_{t+k} \|_2, \tag{14}$$

where $\hat{x}_{t+k}$ and x_{t+k} denote the estimated intention position and the ground-truth position at time step $t + k$, respectively, and K is the prediction horizon. ADE evaluates the overall consistency between inferred intention and realized motion across time.

Final Displacement Error (FDE) (Alahi et al., 2016). FDE measures the Euclidean distance at the final time step, capturing the long-term alignment between the estimated intention and the actual outcome. It is defined as:

$$\mathrm{FDE} = \| \hat{x}_{t+K} - x_{t+K} \|_2. \tag{15}$$

Lower ADE and FDE values indicate that the estimated spatial intention is more consistent with the realized behavior, thereby reflecting higher reliability of the intention modeling and its effectiveness in supporting downstream risk estimation. These metrics also provide an indirect validation of the proposed intention-driven risk modeling framework, by linking the quality of intention estimation to its effectiveness in downstream risk identification.

4 Experiments

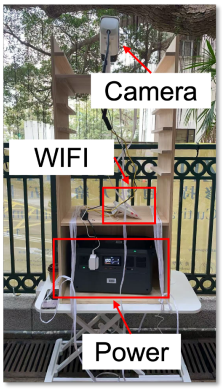
4.1 Experimental Setup

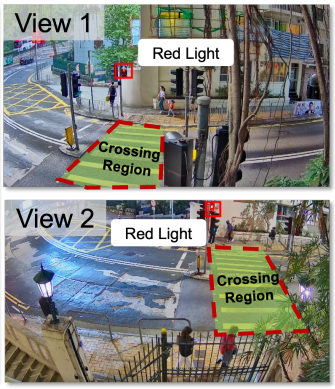
4.1.1 Datasets

This study utilizes a large-scale video dataset collected at a signalized intersection near the University of Hong Kong. This location features high pedestrian density and complex traffic interactions, providing a realistic and challenging benchmark for analyzing urban pedestrian behavior.

As illustrated in **Figure 2** (a)–(b), a single camera was deployed sequentially at two positions at the same signalized intersection, capturing different views of pedestrian movements during separate recording periods. Both deployments used the same acquisition configuration, including a stable power supply, wireless transmission modules, and consistent recording settings. The recorded videos are processed using detection and

tracking algorithms to generate annotated samples, including pedestrian and vehicle bounding boxes with identity labels. In total, the dataset comprises 956 video segments, each approximately 3 minutes long, totaling over 250 GB of footage.





class	X	Y	track_id
2	0.201953	0.597222	1
2	0.375977	0.289583	2
2	0.676562	0.485330	3
5	0.0774902	0.074566	4
0	0.117773	0.156597	5
0	0.532812	0.192882	6
9	0.509375	0.150694	7
0	0.510938	0.228819	8
0	0.285937	0.150608	9
0	0.266797	0.152604	10
0	0.151562	0.163021	11
9	0.464844	0.153646	12
0	0.179004	0.143750	13
0	0.163672	0.149826	14
2	0.0484619	0.154340	15
0	0.417969	0.174957	16
9	0.469141	0.068750	17

(a) Devices (b) Camera Views (c) Data Labeling (d) Trajectory Logs

Figure 2. Overview of the data collection environment and automated preprocessing pipeline.

To ensure data usability and compliance with privacy requirements, a unified preprocessing pipeline is developed, as shown in **Figure 2** (c). A YOLO-based detection and tracking backbone (Redmon et al., 2016) is employed across all stages to maintain consistency in visual perception. The preprocessing includes the following components:

Trajectory Extraction. Pedestrian and vehicle trajectories are extracted using the unified tracking framework. Objects are detected in each frame with category labels, unique track identifiers, and spatial coordinates. The bottom-center of each bounding box is used to approximate the object's location. Consistent track IDs are used to associate detections across frames, forming continuous trajectories, as shown in **Figure 2** (d).

Privacy Protection. All data collection procedures were conducted with appropriate permissions and ethical approval. To protect personal privacy, facial regions of pedestrians are automatically detected and anonymized using Gaussian blurring, removing identifiable features while preserving scene structure.

Traffic Signal Annotation. Traffic signal states are automatically annotated for each frame. Detected traffic light regions are analyzed using RGB color distributions to determine signal states, which are temporally synchronized with trajectory data $\mathbf{L}_t$. This process produces structured sequences combining anonymized visuals, object trajectories, and environmental annotations.

As shown in **Table 1**, the processed dataset contains 30,231 pedestrian trajectories and 36,791 vehicle trajectories. Among these, 1,390 red-light-running events are identified, reflecting the sparsity and complexity of violation behaviors in real-world traffic scenarios.

Table 1. Dataset description.

Metric	Value
Total Pedestrian Trajectories	30,231
Total Vehicle Trajectories	36,791
Confirmed Red-Light-Running	1,390

4.1.2 Implementation Details

The proposed framework is evaluated following a standard trajectory prediction protocol. Models are first pre-trained on the ETH-UCY dataset (Lerner et al., 2007; Pellegrini et al., 2009) to capture general pedestrian motion and interaction patterns, and then evaluated on the collected dataset. The shared bird's-eye view representation across datasets facilitates effective cross-dataset transferability. We adopt an observation horizon of $H = 8$ frames (3.2 seconds) and a prediction horizon of $K = 12$ frames (4.8 seconds), which are consistent with common settings in trajectory prediction literature. For the Intention Prediction module, the number of Monte Carlo samples is set to $N = 1000$ to ensure stable estimation of the intention distribution. The Intention-to-Action Mapping module uses the empirical threshold $\tau = \mu - 2\sigma$, where μ and σ are the sample mean and standard deviation of crossing-region probability scores associated with confirmed violations in an independent calibration set. The same two-standard-deviation rule is applied separately to each model, and the resulting thresholds are fixed during testing. All experiments are conducted using PyTorch on a workstation equipped with a single NVIDIA RTX 4090 GPU (24GB memory).

To examine how different prediction architectures affect intention estimation and downstream red-light-running evaluation, we implement HPDM with five prediction backbones: FlowChain, RNN, LSTM, Transformer, and STGCNN. FlowChain serves as the primary probabilistic predictor as described in Section 3.2.2, and the other backbones are summarized below:

- **RNN** (Elman, 1990) serves as a sequential modeling baseline, capturing short-term temporal dependencies but lacking the capacity to model complex intention evolution.
- **LSTM** (Hochreiter and Schmidhuber, 1997) enhances temporal modeling through memory and gating mechanisms, enabling improved long-range dependency learning.
- **Transformer** (Vaswani et al., 2017) leverages self-attention to capture global temporal dependencies, offering greater flexibility in sequence modeling.
- **STGCNN** (Mohamed et al., 2020) incorporates spatial interaction modeling via graph convolution, capturing both pedestrian interactions and temporal dynamics.

Under the original implementation, RNN, LSTM, and Transformer operate as deterministic predictors, each producing a single future trajectory without an explicit probability distribution. Their outputs therefore do not directly support crossing region probability estimation. To enable a consistent comparison with the stochastic predictors STGCNN and FlowChain within HPDM, we extend these models with probabilistic

prediction heads, yielding HPDM (RNN-GMM), HPDM (LSTM-GMM), and HPDM (Transformer-GMM).

The three variants follow a unified design comprising a historical trajectory encoder and a Gaussian mixture model (GMM) prediction head. Each variant takes normalized historical pedestrian coordinates and consecutive displacements as input. A two-layer RNN, LSTM, or Transformer encoder extracts a 64-dimensional trajectory representation, which is passed to a GMM head with the same architecture across variants. For each future time step, the head predicts a five-component bivariate Gaussian mixture distribution, enabling trajectory sampling and crossing region probability estimation. All three variants are trained end-to-end under identical training settings by minimizing the negative log-likelihood of the ground-truth future positions under the predicted distributions.

4.2 Experimental Analysis

4.2.1 Overall Performance

Quantitative evaluation begins with an assessment of detection performance on the collected dataset. As summarized in **Table 2**, HPDM (FlowChain) achieves the highest accuracy of 96.74%, with the lowest false alarm rate (FAR) of 3.18% and missed detection rate (MDR) of 5.29% among all variants. The results demonstrate that this implementation can effectively distinguish red-light-running behaviors from non-violating behaviors.

Table 2. Quantitative results of different HPDM variants on the collected dataset.

Variants	Accuracy ↑	FAR ↓	MDR ↓
HPDM (RNN-GMM)	92.11%	7.89%	7.59%
HPDM (LSTM-GMM)	92.30%	7.70%	10.13%
HPDM (Transformer-GMM)	92.44%	7.56%	7.59%
HPDM (STGCNN)	95.67%	4.11%	8.20%
HPDM (FlowChain)	96.74%	3.18%	5.29%

To examine performance across different intention prediction models, we compare five HPDM variants within the same framework. HPDM (STGCNN) achieves the second-highest accuracy of 95.67%, with a FAR of 4.11% and an MDR of 8.20%. The RNN-GMM, LSTM-GMM, and Transformer-GMM variants achieve accuracies of 92.11%, 92.30%, and 92.44%, respectively. Although HPDM (FlowChain) performs best, the other variants also achieve competitive performance.

More importantly, these results highlight the flexibility of the proposed framework. The intention prediction module can be replaced without modifying the overall pipeline, demonstrating a plug-and-play property. This suggests that the framework partially decouples intention modeling from downstream risk decision-making, providing a unified mechanism for transforming predicted intentions into behavioral decisions.

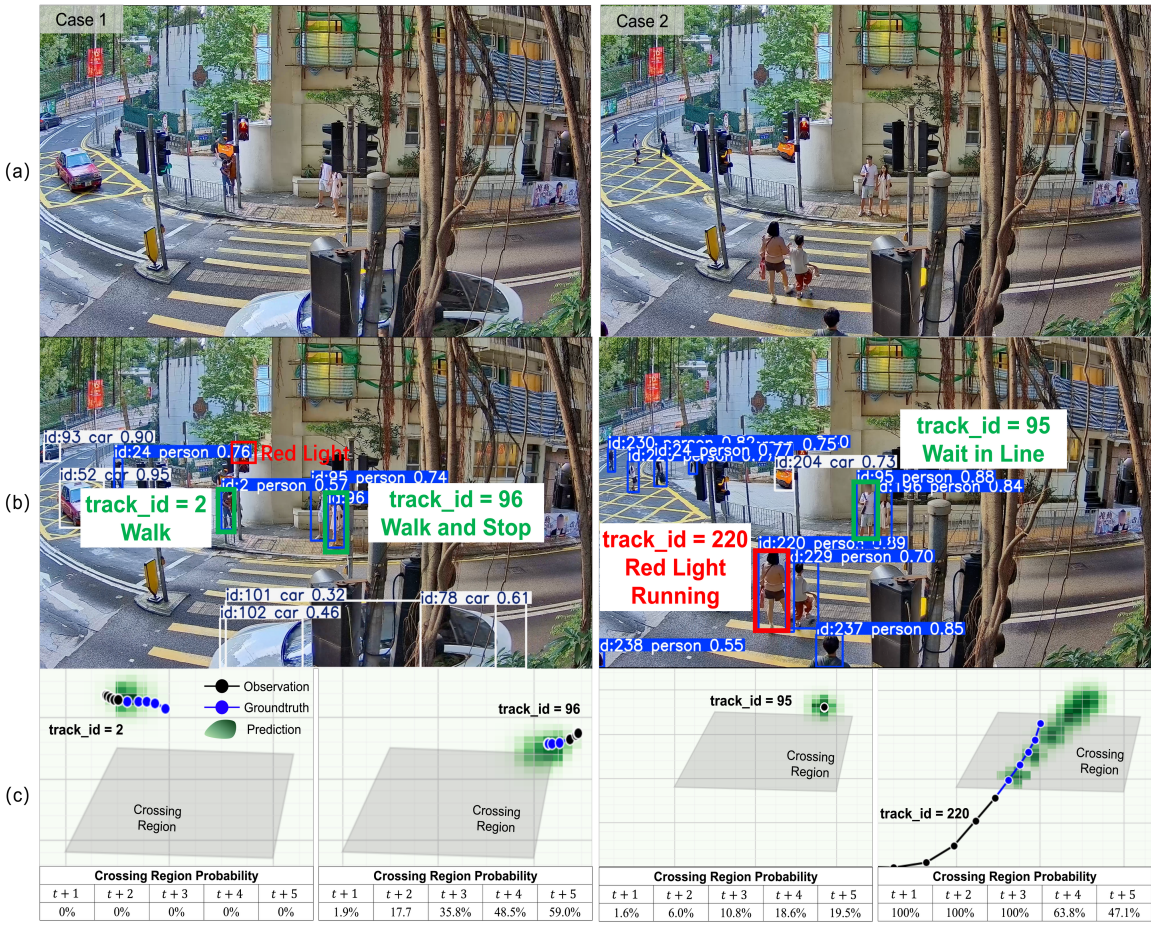


Figure 3. Qualitative visualization of pedestrian intention evolution and red-light-running risk across representative scenarios.

4.2.2 Intention Prediction

Qualitative Analysis. **Figure 3** shows that HPDM (FlowChain) effectively captures diverse pedestrian behaviors and their evolving crossing intentions. In detail, each column corresponds to a real-world case, including detected agents and annotated behaviors. In **Figure 3** (c), **black curves** denote the observed trajectories, **blue curves** represent the ground-truth future trajectories, and **green heatmaps** illustrate the predicted trajectory distributions. The crossing region probability represents the estimated probability that the pedestrian's position lies within the designated crossing region at each future time step.

In compliant scenarios, such as continuous walking along the sidewalk or waiting near the curb, the predicted spatial intention remains outside the crossing region, and the estimated crossing-region probabilities remain low or increase only mildly. In contrast, for pedestrians exhibiting hesitation or red-light-running behavior, the predicted intention progressively shifts toward the crossing area, accompanied by a marked increase in crossing-region probability over future time steps. More importantly, HPDM (FlowChain) is able to distinguish subtle behavioral differences, such as walking-and-stopping patterns

versus actual crossing decisions. This capability arises from its probabilistic modeling of intention evolution, where the spatial intention is represented as a dynamic distribution rather than a single deterministic trajectory. As a result, the predicted crossing region probabilities reasonably reflect the gradual transition from safe to risky behaviors. These observations highlight the model's ability to jointly model spatial intention and violation risk in a coherent and interpretable manner.

Deterministic Models Comparison. Quantitative evaluation is conducted using Average Displacement Error (ADE) and Final Displacement Error (FDE), as summarized in **Table 3**. Sequential models such as RNN and LSTM exhibit relatively large estimation errors (ADE $>$ 1.0 m), indicating limited capability in capturing complex intention dynamics. The Transformer achieves moderate improvements by modeling global temporal dependencies, but still lacks explicit modeling of interaction structures.

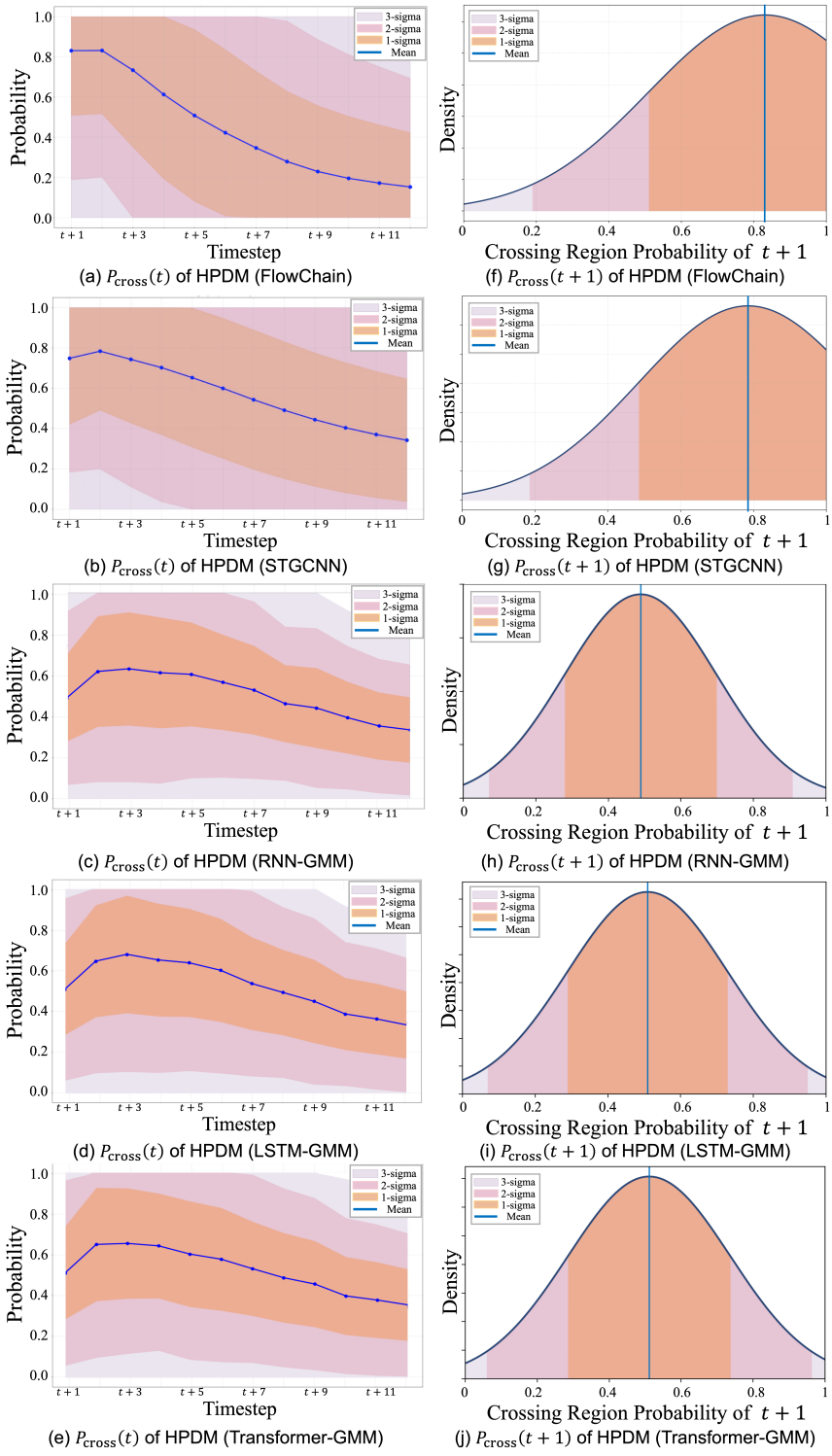
Table 3. Quantitative comparison of different prediction models on the collected dataset. ADE and FDE are reported in meters, and lower indicates better.

Models	Type	ADE ↓	FDE ↓
RNN	Deterministic	1.1994	1.7753
LSTM		1.0551	1.7534
Transformer		1.0361	1.5508
RNN-GMM	Stochastic	0.9053	1.4429
LSTM-GMM		0.8974	1.4470
Transformer-GMM		0.8803	1.2340
STGCNN		0.5457	0.9098
FlowChain		0.5272	0.8157

Probabilistic Extensions. Adding a GMM prediction head improves both metrics for all three backbones. RNN-GMM reduces ADE and FDE to 0.9053 m and 1.4429 m, while LSTM-GMM achieves 0.8974 m and 1.4470 m. Transformer-GMM performs best among these extensions, achieving an ADE of 0.8803 m and an FDE of 1.2340 m. These consistent improvements show that the GMM-based implementations combine probabilistic output representation with lower displacement errors. Their predicted spatial distributions also enable crossing region probability estimation, connecting trajectory prediction with the continuous behavioral measurement required by HPDM.

Effectiveness of Flow-Based Intention Modeling. Compared with GMM-based variants, STGCNN performs better, with an ADE of 0.5457 m and an FDE of 0.9098 m. FlowChain further reduces these errors to 0.5272 m and 0.8157 m, respectively, achieving the best performance. This improvement can be attributed to its ability to model the distribution of future spatial intentions through continuous, invertible transformations. Unlike deterministic or single-path prediction models, the flow-based formulation captures the multimodal nature of pedestrian behavior and enables smooth probability transitions

across different potential intentions. Furthermore, it supports stable uncertainty propagation over time, leading to more accurate and consistent intention estimation. These results indicate that incorporating flow-based intention modeling enhances the quality of spatial intention estimation, providing a more reliable foundation for subsequent risk quantification within the proposed prediction-driven measurement framework.



(a) $P_{\text{cross}}(t)$ of HPDM (FlowChain)
(f) $P_{\text{cross}}(t+1)$ of HPDM (FlowChain)
(b) $P_{\text{cross}}(t)$ of HPDM (STGCNN)
(g) $P_{\text{cross}}(t+1)$ of HPDM (STGCNN)
(c) $P_{\text{cross}}(t)$ of HPDM (RNN-GMM)
(h) $P_{\text{cross}}(t+1)$ of HPDM (RNN-GMM)
(d) $P_{\text{cross}}(t)$ of HPDM (LSTM-GMM)
(i) $P_{\text{cross}}(t+1)$ of HPDM (LSTM-GMM)
(e) $P_{\text{cross}}(t)$ of HPDM (Transformer-GMM)
(j) $P_{\text{cross}}(t+1)$ of HPDM (Transformer-GMM)

Figure 4. Comparative analysis of the intention-to-action mapping across different HPDM variants.

4.2.3 Intention-to-Action Mapping

Figure 4 compares crossing region probability dynamics and onset-aligned score distributions across five HPDM variants. Subfigures (a)–(e) show the temporal profiles from $t + 1$ to $t + 12$, while subfigures (f)–(j) show the corresponding distributions at $t + 1$. The analysis aligns confirmed violations by their observed onset to examine how predicted crossing propensity relates to realized crossing behavior.

Probabilistic Intention Dynamics. The five variants exhibit a common temporal pattern: crossing region probabilities reach relatively high levels near the onset of violation and subsequently decline. HPDM (FlowChain) starts at approximately 0.83 and decreases to approximately 0.15 by $t + 12$. HPDM (STGCNN) reaches an early peak of approximately 0.78 before declining to approximately 0.34. The three GMM-based variants start near 0.50, rise to approximately 0.63–0.68 within the first few steps, and then decrease to about 0.33-0.35. Thus, the variants differ in probability magnitude and peak timing but consistently capture an early concentration of crossing propensity followed by a decline. These profiles describe predicted occupancy of the crossing region. Their decrease reflects the evolving spatial relationship between predicted pedestrian positions and the region.

Statistical Intention-to-Action Mapping. Subfigures (f)–(j) summarize the crossing region probability distributions at the initial violation-aligned time step, $t + 1$, using Gaussian approximations restricted to the probability range $[0, 1]$. These curves illustrate differences in distribution shape and location across models; threshold calculation uses the empirical scores directly. HPDM (FlowChain) and HPDM (STGCNN) produce distributions centered at higher scores, approximately 0.83 and 0.78, respectively, whereas the three GMM-based variants are centered near 0.50. These differences motivate model-specific threshold calibration rather than a shared numerical threshold.

For each variant, the decision threshold is calculated as $\tau = \mu - 2\sigma$, where μ and σ are the empirical mean and standard deviation of the onset-aligned crossing region probability scores in the calibration data. This empirical two-standard-deviation rule provides a common calibration procedure while allowing model-specific thresholds to accommodate differences in score distributions. The resulting mapping converts continuous crossing propensity scores into discrete violation decisions when required, while retaining the full probability profiles for behavioral analysis. Its application across the five variants demonstrates compatibility with different prediction architectures.

Context-Aware Intention-to-Action Mapping. Further analysis reveals that surrounding traffic plays an important role in shaping pedestrian intention. **Figure 5** compares scenarios with and without vehicle interaction. In detail, **Figure 5** (a) illustrates

the scenario before the approaching vehicle enters the crossing region. The pedestrian exhibits a clear forward-moving intention, with predicted crossing region probabilities rising to nearly 40% in the near future. As shown in **Figure 5** (b), the predicted intention distribution collapses toward the current position, and the crossing region probability drops to a negligible level across all corresponding time steps. This strong inhibitory effect reflects the influence of traffic constraints on pedestrian decision-making. Based on this observation, we incorporate a context-aware constraint into the mapping process: pedestrians are conservatively treated as low-risk when a vehicle enters the crossing region. This design is grounded in observed behavioral patterns and effectively reduces false positive predictions.

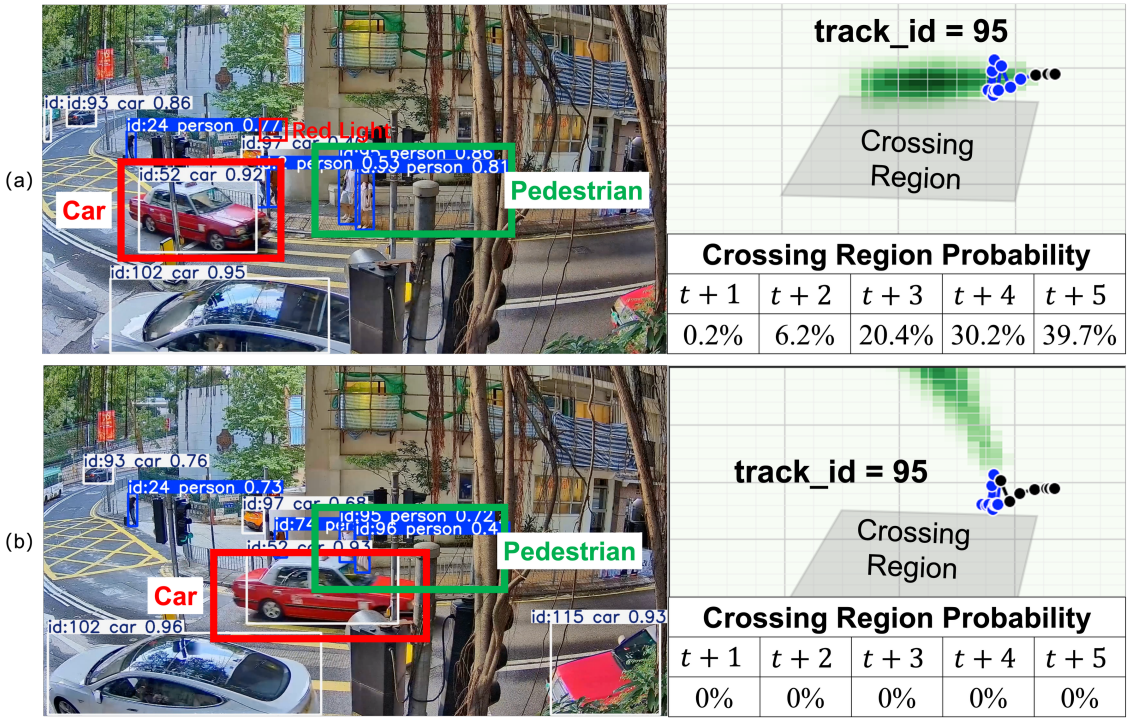


Figure 5. Impact of approaching vehicles on pedestrian crossing intention. The legend is as the same as that in Figure 3. For clarity, only the crossing region probabilities for the first five future frames ($t + 1$ to $t + 5$) are displayed.

4.3 Discussion

4.3.1 Performance on a Balanced Test Set

Red-light-running events account for approximately 4.6% of pedestrian trajectories in the collected dataset. To examine whether the performance reported on the full dataset is influenced by the predominance of non-violating trajectories, we conduct an additional evaluation on a balanced test set containing 417 violating and 417 non-violating pedestrian trajectories. We also report confusion matrices, precision–recall (PR) curves, and PR-AUC to assess class-specific errors and the precision–recall trade-off across decision thresholds.

As reported in **Table 4**, HPDM (FlowChain) achieves the best performance, with an accuracy of 95.80%, a FAR of 3.12%, and an MDR of 5.28%. Its FAR and MDR closely match those obtained on the full collected dataset. Meanwhile, HPDM (STGCNN) similarly maintains comparable error rates, with a FAR of 4.08% and an MDR of 8.15%, compared with 4.11% and 8.20% on the full dataset. The three GMM-based variants achieve accuracies of 91.13%–92.33%, remaining close to their full-dataset results of 92.11%–92.44%. These results indicate that the HPDM variants retain good identification performance on the balanced test set.

Table 4. Quantitative results of different HPDM variants on a balanced test set.

Variants	Accuracy ↑	FAR ↓	MDR ↓	PR-AUC ↑
HPDM (RNN-GMM)	91.61%	7.91%	8.87%	0.9100
HPDM (LSTM-GMM)	91.13%	7.67%	10.07%	0.9155
HPDM (Transformer-GMM)	92.33%	7.67%	7.67%	0.9292
HPDM (STGCNN)	93.88%	4.08%	8.15%	0.9411
HPDM (FlowChain)	95.80%	3.12%	5.28%	0.9643

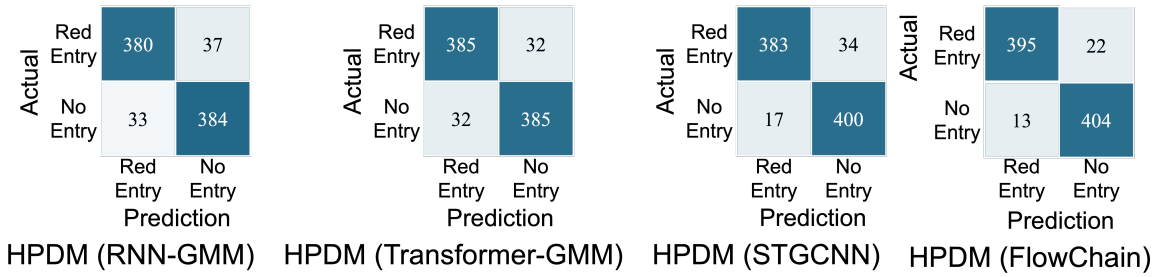


Figure 6. Confusion matrices of different HPDM variants.

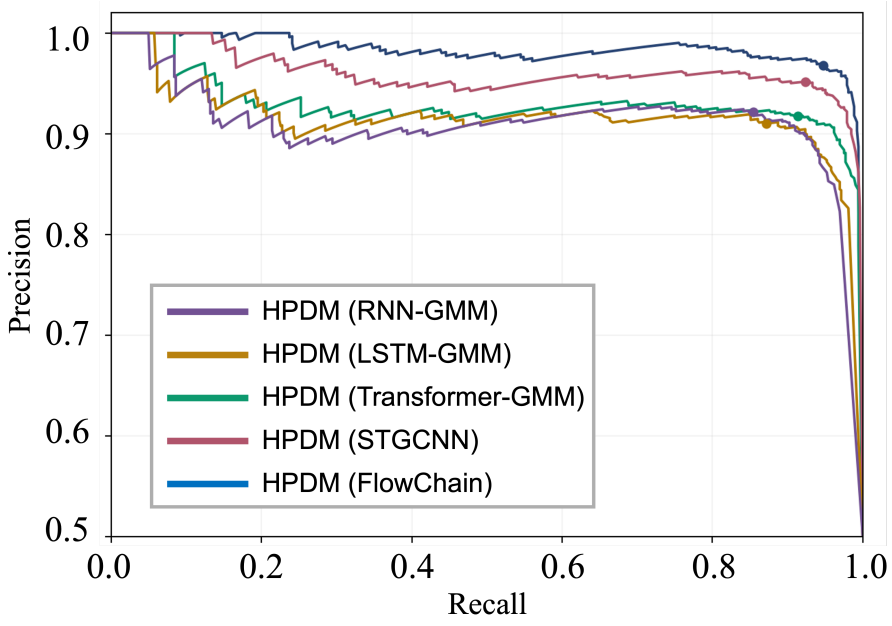


Figure 7. PR curves of different HPDM variants.

The confusion matrices in **Figure 6** illustrate the errors at the calibrated decision thresholds, while the PR curves in **Figure 7** assess performance across varying thresholds. PR-AUC ranges from 0.9100 to 0.9643, with HPDM (FlowChain) attaining the highest value of 0.9643, followed by HPDM (STGCNN) at 0.9411. Together, these results show that HPDM maintains good violation identification performance when the two classes are equally represented.

4.3.2 Effectiveness of Prediction Lead Time

The value of proactive monitoring depends on how far in advance potential violations can be identified with sufficient reliability. Although HPDM generates probability profiles over a 12-step horizon spanning 4.8 s, prediction performance can vary across future time steps. We therefore extend the next-step evaluation to five prediction horizons, $t + 1$, $t + 3$, $t + 6$, $t + 9$, and $t + 12$, corresponding to lead times of 0.4, 1.2, 2.4, 3.6, and 4.8 s. The evaluation uses eligible trajectories from the balanced test set, with thresholds calibrated separately for each model and horizon on an independent calibration set and fixed during testing. We repeat the evaluation 50 times with different random sampling realizations and report the results in Table 5.

Table 5. Red-light-running prediction accuracy (%) of different HPDM variants at different prediction lead times on a balanced test set (mean ± standard deviation).

Variants	$t + 1$	$t + 3$	$t + 6$	$t + 9$	$t + 12$
HPDM (RNN-GMM)	92.75±0.41	84.20±0.63	58.31±2.98	43.82±1.68	27.33±1.62
HPDM (LSTM-GMM)	91.85±0.33	81.61±0.94	60.85±2.18	43.05±1.57	25.31±2.92
HPDM (Transformer-GMM)	92.59±0.50	85.13±0.79	61.13±1.35	45.58±1.53	23.97±2.74
HPDM (STGCNN)	95.28±0.28	87.34±1.81	67.25±1.65	47.75±1.66	22.39±2.97
HPDM (FlowChain)	96.46±0.31	85.66±3.64	64.95±1.84	47.22±1.12	29.81±2.89

As shown in **Table 5**, prediction accuracy decreases markedly as lead time increases. HPDM (FlowChain) achieves the highest mean accuracy of 96.46% at 0.4 s, while HPDM (STGCNN) performs best at 1.2 s, reaching 87.34%. At 2.4 s, the highest mean accuracy decreases to 67.25%, followed by 47.75% at 3.6 s and 29.81% at 4.8 s. All HPDM variants exhibit this trend, indicating that longer prediction horizons substantially increase the difficulty of identifying future violations. At longer lead times, observable motion cues may be less informative about subsequent crossing behavior, while vehicle movements and signal conditions can continue to evolve.

These findings identify 0.4–1.2 s as the most promising range for anticipatory monitoring under the evaluated conditions, with substantially weaker performance beyond this range. HPDM therefore provides its strongest predictive support for frequent risk updates and timely warnings close to a potential violation.

4.3.3 Limitations and Future Work

The trajectory data record what pedestrians do, but do not directly reveal why they decide to cross. HPDM therefore uses crossing region probability as an indicator of crossing propensity rather than a direct measure of psychological intention. Pedestrians with similar observed trajectories may act differently because of distraction, physical ability, perceived risk, or changes in weather and road conditions. Tracking errors and occlusion can also affect the predictions. The results at longer horizons show that historical motion alone provides limited information about actions several seconds ahead. Future work will include additional cues, such as body orientation and head movements, together with environmental observations. Where feasible, interviews or participant reports could help examine how these motion-based indicators relate to stated intentions.

The data were collected under sunny conditions at a single signalized intersection near the University of Hong Kong, using two camera positions during separate recording periods. Pedestrian behavior, signal timing, crossing layout, and traffic composition may differ elsewhere, affecting both prediction performance and the decision thresholds. Testing different prediction backbones demonstrates the flexibility of HPDM, but its performance at other intersections remains to be assessed. We will extend data collection to locations with different layouts and operating conditions, and test the framework on intersections excluded from training. These experiments will establish when local recalibration or model adaptation is needed. Field trials will then examine whether warnings can be delivered early enough to influence behavior and improve safety.

5 Conclusion

Pedestrian red-light running remains a prevalent and safety-critical issue in urban traffic systems, yet existing studies predominantly rely on outcome-based detection paradigms that provide limited insight into how risky behaviors emerge and evolve. To address this limitation, this study proposes a hierarchical prediction-driven measurement framework (HPDM) that connects probabilistic trajectory prediction with continuous behavioral measurement and violation identification.

The framework comprises two connected stages: intention prediction and intention-to-action mapping. Given a scene representation integrating historical pedestrian trajectories, traffic signal states, crossing region geometry, and surrounding vehicle conditions, the first stage estimates future trajectory distributions and computes temporally resolved crossing region probabilities. These probabilities serve as continuous indicators of crossing propensity under red-light conditions and as a common intermediate representation connecting the two stages. The mapping stage applies an empirically calibrated threshold to generate discrete violation decisions when required, while retaining the full probability profiles for behavioral analysis. The contribution lies in this modular measurement pipeline, with prediction models serving as backbone components and thresholding as an interpretable decision rule.

Experimental results on a real-world dataset demonstrate that the proposed framework can effectively distinguish red-light-running behaviors from non-violating behaviors. HPDM (FlowChain) achieves an accuracy of 96.74% on the collected dataset and 95.80% on a balanced test set, with a PR-AUC of 0.9643 on the latter. Comparisons across five prediction backbones demonstrate that alternative implementations also provide good identification performance. Their integration through a common probabilistic output interface highlights the framework's flexibility and partially decouples prediction architecture from downstream behavioral measurement. Evaluation across prediction horizons further identifies shorter lead times, particularly 0.4–1.2 s, as the most promising range for anticipatory monitoring under the evaluated conditions.

Overall, this study shifts the analysis of pedestrian red-light running from retrospective violation detection to continuous, prediction-driven measurement of crossing propensity, offering a new perspective for understanding safety-relevant pedestrian behavior in complex urban environments. Future work will extend the framework to more complex multi-agent traffic scenarios involving richer interaction patterns and heterogeneous road users, evaluate cross-scene transferability, and examine how prediction reliability and system latency jointly determine usable warning time and practical intervention benefits.

Ethical Statement

All data collection procedures in this study were conducted in compliance with applicable regulations and ethical standards. The study protocol was reviewed and approved by the Human Research Ethics Committee of The University of Hong Kong (HREC reference number: EA250504).

CRediT Authorship Contribution Statement

Beihao Xia: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Validation. Bo Liu: Writing – review & editing, Software, Visualization. Heng Li: Writing – review & editing, Formal analysis. Wei Yu: Writing – review & editing, Investigation. Si Qiao: Writing – review & editing, Supervision, Project administration, Conceptualization, Methodology. Qinmu Peng: Writing – review & editing, Funding acquisition. Zhu Xiao: Supervision, Resources. Xinge You: Supervision. Anthony Gar-On Yeh: Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Data Availability

Data will be made available on request.

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